Pulleys and Strings

Answer key

Conventions used throughout

• **Forces:** F_g = gravity ($F_g = mg$), so $F_{ga} = m_a g$ and $F_{gb} = m_b g$. T = tension, F_N = normal force, F_s = static friction (largest value $\mu_s F_N$), F_f = friction on a sliding object ($F_f = \mu F_N$).

• **Strings and pulleys:** a light string over a frictionless pulley has the same tension T all along it. A string that doesn't stretch makes every mass on it move with the same size of acceleration a.

• **Signs (from the lesson):** each mass gets its own + direction, pointing the way that mass accelerates (or would move). Then a is positive for every mass. Forces are written as sizes, with + or − in the equation for their direction, as the worksheet asks.

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

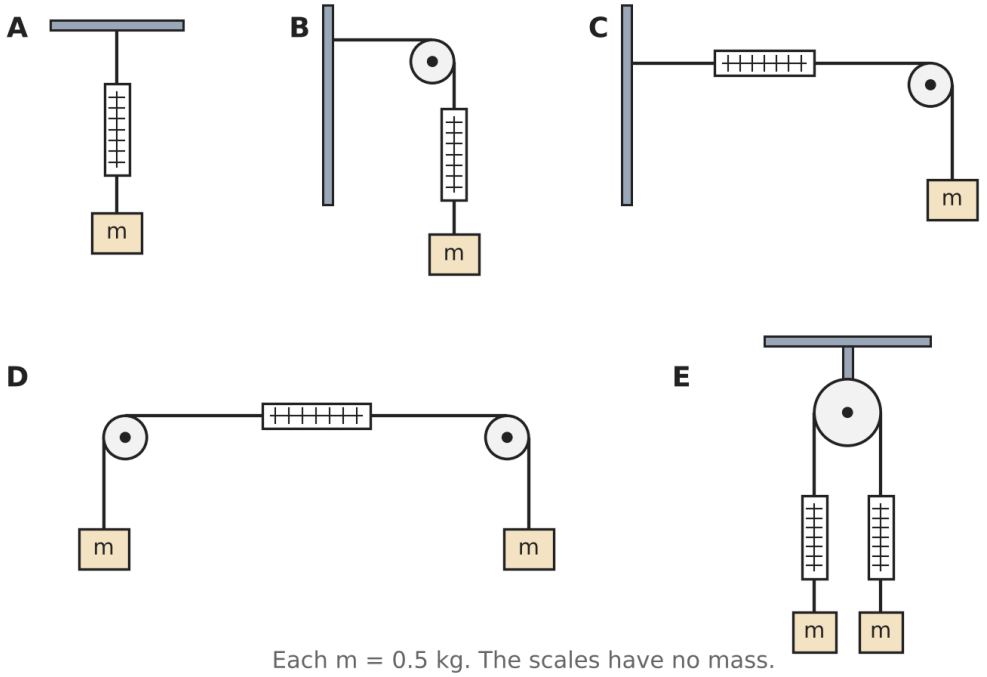
TENSION & PULLEYS

Spring scales in five set-ups

Suppose you were to hang a mass of 0.5 kg in the various configurations shown below. You may assume the scales have no mass.

1. Draw a free body diagram for the spring scale itself in each scenario.

The five set-ups (as on the worksheet)



Each m = 0.5 kg. The scales have no mass.

Step 1 – How hard each string pulls

Each hanging mass is at rest, so the string holding it pulls up exactly as hard as gravity pulls down:

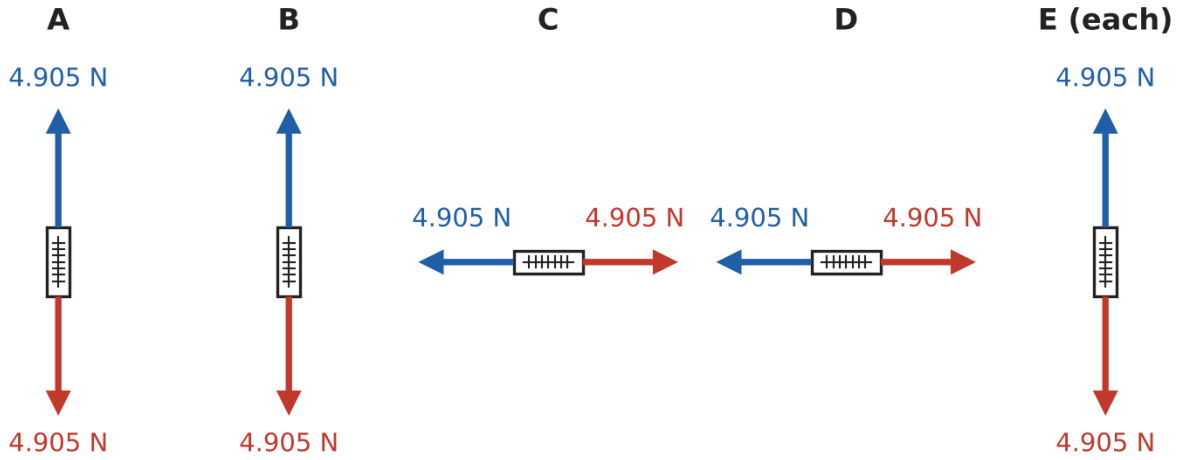
Formula: $F_T = F_g = mg$

$F_T = (0.5\ \text{kg})(9.81\ \text{m/s}^2) = 4.905\ \text{N}$

Step 2 – The forces on each scale

A light string over a frictionless pulley has the same tension all along it, so every string tied to a scale pulls with 4.905 N. The scale has no mass, so no gravity acts on it. It doesn't accelerate, so the pull at one end must be equal and opposite to the pull at the other end:

FBD of the spring scale in each set-up



Every arrow is a tension force $\vec{F}_T$ from a string attached to the scale.

Set-up	What pulls on the two ends of the scale
A	The string to the ceiling pulls up; the mass's string pulls down.
B	The string over the pulley (tied to the wall) pulls up; the mass's string pulls down.
C	The string tied to the wall pulls left; the string over the pulley to the mass pulls right.
D	Each end has a string over a pulley to a hanging mass. One mass does the job the wall did in C, so the scale still feels 4.905 N at each end, not double.
E	On each scale, the string over the pulley pulls up and that scale's mass pulls down.

FINAL ANSWER

Every scale has two equal and opposite tension forces of 4.905 N, one at each end

So every scale reads mg = 4.905 N ≈ 5 N, in all five set-ups (even D)

ATWOOD MACHINE

Equal and unequal masses

Typical pulley problems will consist of an Atwood Machine.

Atwood Machine – consists of two weights tied together and suspended over a pulley as the photo to the right depicts.

1) If the masses are equal, if we were to give one a gentle push to a mass and release it, what type of motion will occur **after** it is released?

2) Suppose that m_b is greater than m_a. Will mass A accelerate? If so, in what direction? Will mass B accelerate? If so, in what direction? How will those accelerations compare in magnitude? Explain.

1) If the masses are equal, if we were to give one a gentle push to a mass and release it, what type of motion will occur **after** it is released?

With equal masses, gravity pulls each side down equally hard, so after the push there is no net force on the system:

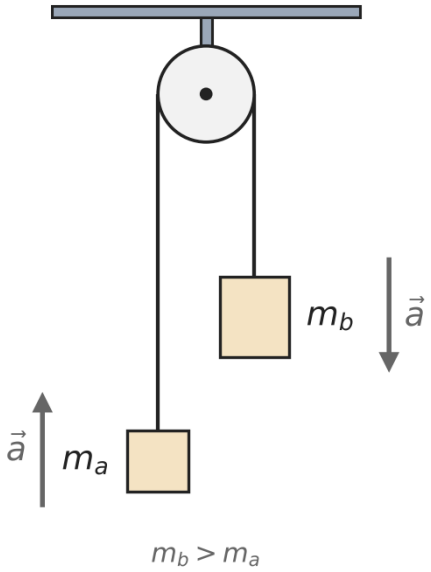
$$F_{net} = F_{gb} - F_{ga} = mg - mg = 0 \quad \Rightarrow \quad a = 0$$

By Newton's first law, the masses keep the velocity the push gave them. One mass moves up and the other moves down, both at a constant speed (ignoring friction in the pulley).

2) Suppose that m_b is greater than m_a. Will mass A accelerate? ... How will those accelerations compare in magnitude? Explain.

Yes, both accelerate. B is heavier, so gravity pulls B's side down harder than A's side: B speeds up **downward** and pulls A **upward**. The string doesn't stretch, so when B moves down 1 cm, A moves up 1 cm in the same time. They always have the same speed, so their accelerations have the **same magnitude** (in opposite directions).

The Atwood machine with $m_b > m_a$



FINAL ANSWER

1) Uniform motion: each mass keeps a constant velocity (a = 0)

2) A accelerates up, B accelerates down, with the same magnitude (the string doesn't stretch)

FBDs and symbols

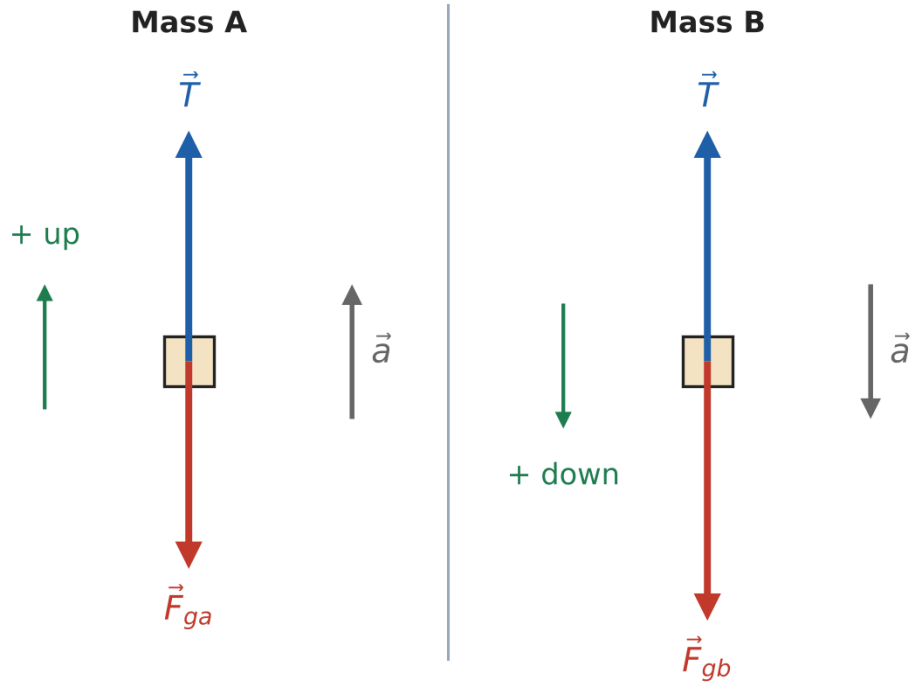
3) Draw a FBD for mass A and for mass B.($m_b > m_a$). Indicate the acceleration of each mass and **choose your sign conventions so they align with the acceleration of the mass**! (positive will be different for each mass)

4) For the FBDs, do you need to use a different symbol for the magnitude of the acceleration of mass A and mass B (i.e. a_a and a_b?) What about for the forces of tension and gravity acting on each mass? Explain.

3) Draw a FBD for mass A and for mass B. Indicate the acceleration of each mass and choose your sign conventions so they align with the acceleration of the mass!

Two forces act on each mass: tension up and gravity down. A accelerates up, so + is up for A and T is longer than F_{ga}. B accelerates down, so + is down for B and F_{gb} is longer than T. (Drawn to scale for the masses in 5d.)

FBDs of mass A and mass B



4) Do you need to use a different symbol for the magnitude of the acceleration of mass A and mass B? What about for the forces of tension and gravity acting on each mass? Explain.

Quantity	Same or different?	Why
Acceleration	Same symbol, a	The string doesn't stretch, so both sizes are equal. With + along each mass's own motion, a is positive for both.
Tension	Same symbol, T	It is one light string over a frictionless pulley, so it pulls up on A and on B equally hard.
Gravity	Different: F_{ga} and F_{gb}	F_g = mg and the masses are different, so the two weights are different.

FINAL ANSWER

3) A: T up, F_{ga} down, + up. B: T up, F_{gb} down, + down (FBDs above)

4) One a and one T for both masses; gravity needs two symbols, F_{ga} and F_{gb}

ATWOOD MACHINE

The system approach

5) Now we must realize that both masses accelerate **together, as a system**.

a. What force(s) act to move the system forward?

b. What force(s) act to oppose the motion of the system?

c. Write an scalar equation for the net force on the system:

d. If m_a = 0.200 kg and m_b = 0.250 kg, determine the acceleration of the system:

"Forward" for the system means the way it moves: B down and A up.

a. What force(s) act to move the system forward?

Gravity on B, F_{gb}. It pulls B down, which is the way the system moves.

b. What force(s) act to oppose the motion of the system?

Gravity on A, F_{ga}. It pulls A down, but A moves up. (Tension is inside the system: it pulls A forward and B backward by the same amount, so it cancels.)

c. Write an scalar equation for the net force on the system:

$$F_{net} = F_{gb} - F_{ga}$$

d. If m_a = 0.200 kg and m_b = 0.250 kg, determine the acceleration of the system:

GIVEN

$m_a = 0.200 \text{ kg}$ $m_b = 0.250 \text{ kg}$ $a = ?$

Step 1 – The two weights

Formula: $F_g = mg$

$F_{ga} = (0.200)(9.81) = 1.962 \text{ N}$ $F_{gb} = (0.250)(9.81) = 2.4525 \text{ N}$

Step 2 – Net force and acceleration

Use the equation from (c). The system's mass is both masses together:

$$F_{net} = F_{gb} - F_{ga} = 2.4525 - 1.962 = 0.4905 \text{ N}$$

Formula: $F_{net} = ma$

Divide both sides by m: $a = \dfrac{F_{net}}{m_a + m_b}$

$$a = \frac{0.4905 \text{ N}}{0.200 \text{ kg} + 0.250 \text{ kg}} = \frac{0.4905}{0.450} = 1.090 \text{ m/s}^2$$

FINAL ANSWER

a. F_{gb} (gravity on mass B)

b. F_{ga} (gravity on mass A)

c. $F_{net} = F_{gb} - F_{ga}$

d. a = 1.09 m/s^2 (A up, B down)

ALTERNATIVE METHOD

Calculate acceleration with two equations

1. Go back to your **FBD** of **mass A**. Use **Newton's 2nd Law** to write down an equation that relates the forces acting on **mass A** to the **acceleration** of **mass A**. Write the forces as scalars and use + or – to indicate direction. (Don't solve for *a*.)

2. Use **Newton's 2nd Law** to write down a **scalar equation** that relates the forces acting on **mass B** to the acceleration of **mass B**.

3. Algebraically eliminate ***T*** from the above two equations and solve for the acceleration of the masses.

4. Use an equation from question **8**. or **9**. to solve for the magnitude of the force of tension. How does this value compare with F_{ga} and F_{gb}? Explain why.

1. Newton's 2nd Law for mass A (+ up)

T points up (+) and F_{ga} points down (−):

Formula: $F_{net} = ma$

Mass A: $T - F_{ga} = m_a a$

2. Newton's 2nd Law for mass B (+ down)

F_{gb} points down (+) and T points up (−):

Mass B: $F_{gb} - T = m_b a$

3. Algebraically eliminate T from the above two equations and solve for the acceleration of the masses.

Put in the numbers from 5d. T is added in one equation and subtracted in the other, so adding the two equations cancels it:

Mass A: $T - 1.962 = 0.200\,a$

Mass B: $2.4525 - T = 0.250\,a$

Add the equations (the T's cancel):

$$2.4525 - 1.962 = 0.200\,a + 0.250\,a$$

Simplify both sides: $0.4905 = 0.450\,a$

Divide both sides by 0.450 : $a = 1.090\ \text{m/s}^2$

The same answer as the system method in 5d.

4. Use an equation from question 8. or 9. to solve for the magnitude of the force of tension. How does this value compare with F_{ga} and F_{gb}? Explain why.

"Question 8. or 9." means the two equations from 1. and 2. above. Use mass A's equation:

Mass A: $T - F_{ga} = m_a a$

Add F_{ga} to both sides: $T = F_{ga} + m_a a$

$T = 1.962 + (0.200)(1.090) = 1.962 + 0.218 = 2.180 \text{ N}$

Compare: $F_{ga} = 1.962$ N < T = 2.180 N < $F_{gb} = 2.4525$ N. A speeds up upward, so the upward pull on it (T) must be bigger than its weight. B speeds up downward, so its weight must be bigger than the upward pull (T). T has to be between the two weights.

FINAL ANSWER

1. $T - F_{ga} = m_a a$

2. $F_{gb} - T = m_b a$

3. $a = 1.09 \text{ m/s}^2$

4. T = 2.18 N, which is between F_{ga} (1.96 N) and F_{gb} (2.45 N)

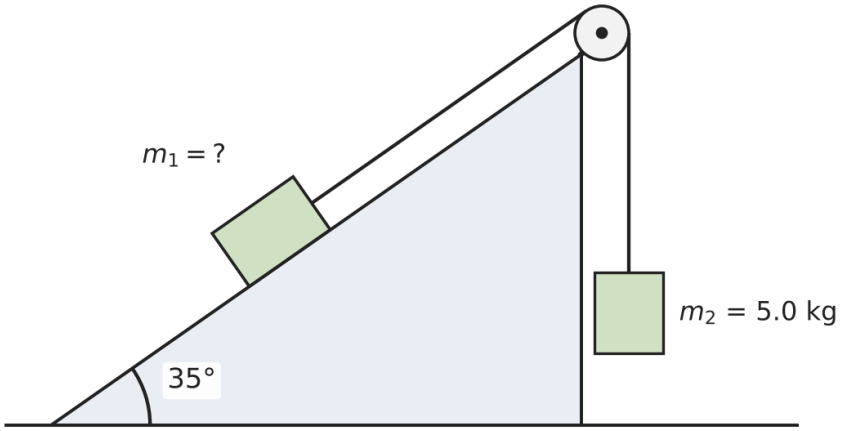
Check: mass B's equation gives the same tension: $T = F_{gb} - m_b a = 2.4525 - (0.250)(1.090) = 2.180$ N.

EXAMPLE 1

Largest mass that stays at rest

Example 1) The figure below shows two blocks joined with a rope that runs over a pulley. The mass of $\mathbf{m_2}$ is **5.0kg**, and the incline is **35°**. The coefficient of static friction between $\mathbf{m_1}$ and the inclined plane is **0.25**. Determine the largest mass for $\mathbf{m_1}$ such that both blocks remain at **rest**.

Picture the situation



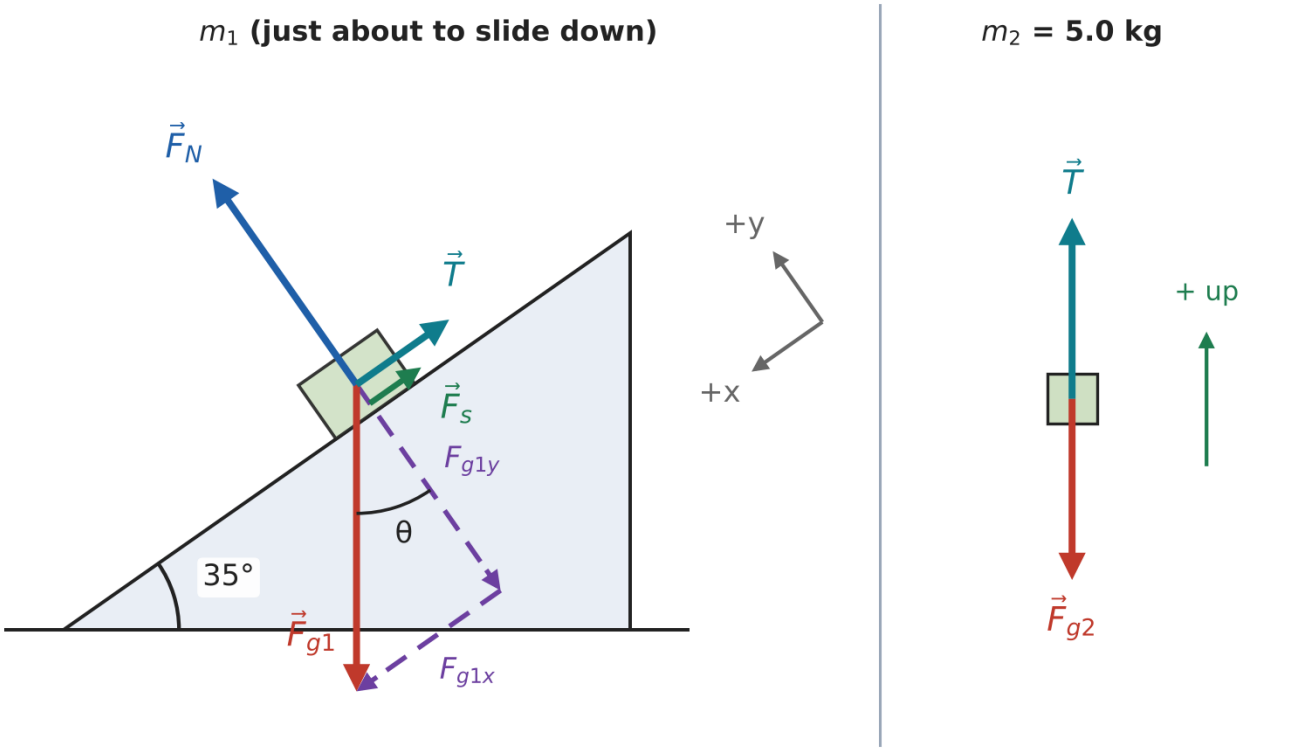
$\mu_s = 0.25$ between m_1 and the ramp

m_2 pulls m_1 up the ramp through the rope. If m_1 is heavy enough, it slides down the ramp and pulls m_2 up.

Step 1 – What "largest m_1" means

As m_1 gets bigger, gravity pulls it down the ramp harder. The rope and static friction both pull it up the ramp. At the largest m_1, it is just about to slide down, so static friction points up the ramp at its largest value, $F_s = \mu_s F_N$. Take + down the ramp for m_1 and + up for m_2 (the way each would move).

FBDs (m_1: +x down the ramp, +y away from the ramp)



GIVEN

$m_2 = 5.0$ kg	$\theta = 35°$	$\mu_s = 0.25$
$a = 0$	$T = ?$	$m_1 = ?$

Step 2 – Tension from m_2

m_2 stays at rest, so the rope holds up its whole weight:

Formula: $F_{net} = ma$

Mass 2, at rest: $T - F_{g2} = 0$

Add F_{g2} to both sides: $T = m_2 g = (5.0)(9.81) = 49.05$ N

Step 3 – Gravity components and normal force on m_1

F_{g1x} (along the ramp) is opposite θ, so it uses sine. F_{g1y} (into the ramp) is next to θ, so it uses cosine. The normal force balances F_{g1y}:

$$F_{g1x} = m_1 g \sin\theta = m_1(9.81)\sin 35° = 5.627\, m_1$$

$$F_{g1y} = m_1 g \cos\theta = m_1(9.81)\cos 35° = 8.036\, m_1$$

y direction, at rest: $F_N - F_{g1y} = 0 \quad \Rightarrow \quad F_N = 8.036\, m_1$

Step 4 – Largest static friction

Formula: $F_s = \mu_s F_N$

$$F_s = (0.25)(8.036\, m_1) = 2.009\, m_1$$

Step 5 – Balance m_1 along the ramp and solve

F_{g1x} points down the ramp (+). T and F_s point up the ramp (−):

Formula: $F_{net} = ma$

Mass 1, at rest: $F_{g1x} - T - F_s = 0$

Put in the expressions: $5.627\, m_1 - 49.05 - 2.009\, m_1 = 0$

Combine the m_1 terms: $3.618\, m_1 - 49.05 = 0$

Add 49.05 to both sides: $3.618\, m_1 = 49.05$

Divide both sides by 3.618 : $m_1 = \dfrac{49.05}{3.618} = 13.56$ kg

Any heavier m_1 would need more friction than the ramp can give, and m_1 would slide.

FINAL ANSWER

Largest m_1 = 14 kg

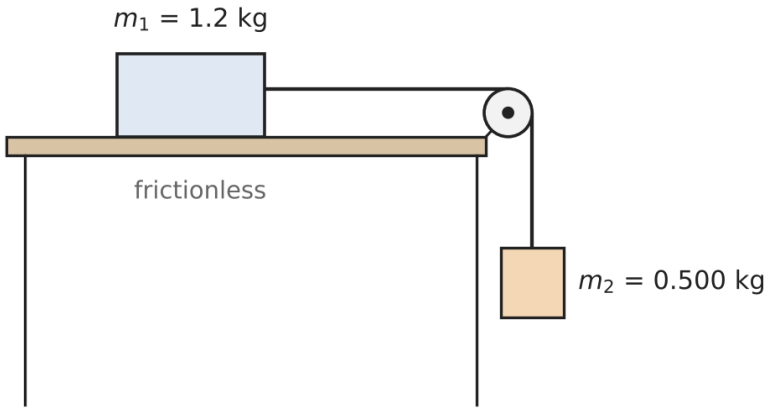
Check: with m_1 = 13.56 kg: F_{g1x} = 76.29 N, and T + F_s = 49.05 + 27.24 = 76.29 N.

TRY THESE 1

Block on a frictionless table

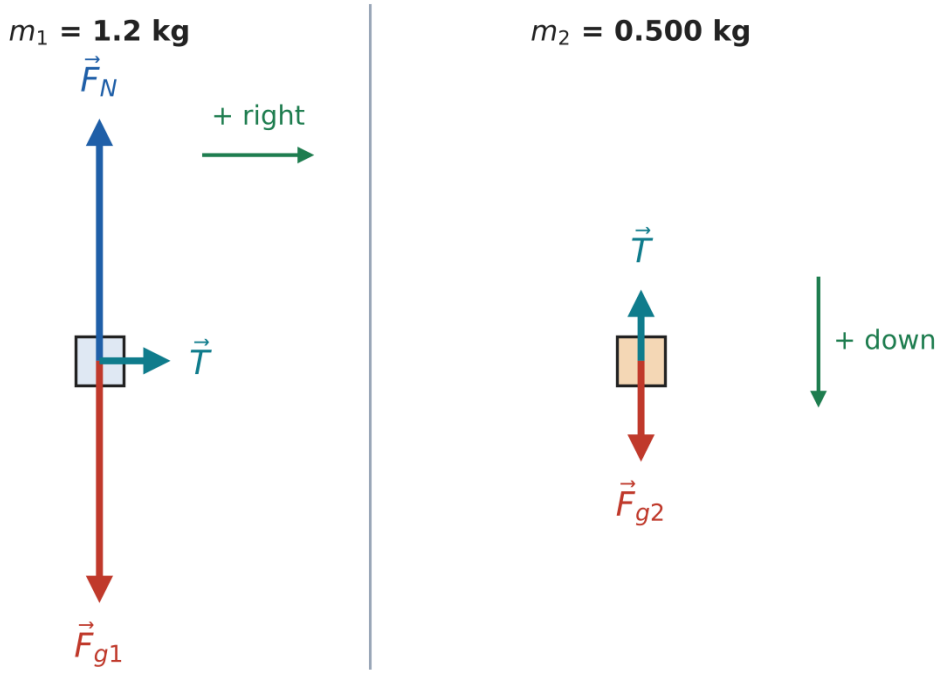
1. In the diagram below, m_1 = 1.2 kg and m_2 = 0.500 kg. The two masses are connected by an inextensible string and m_1 slides along a frictionless surface. Find the acceleration of the system and the string tension. (2.9 m/s^2 3.5 N)

Picture the situation



m_2 falls and drags m_1 along the table toward the pulley.

FBDs (+ for each mass is the way it moves)



Step 1 – Acceleration (system method)

On m_1, F_N and F_{g1} cancel and there is no friction. Tension is inside the system. So the only force driving the system is gravity on m_2, and nothing opposes it:

$$F_{net} = F_{g2} = m_2 g = (0.500)(9.81) = 4.905 \text{ N}$$

$$\text{Formula: } F_{net} = ma \quad \Rightarrow \quad a = \frac{F_{net}}{m_1 + m_2}$$

$$a = \frac{4.905 \text{ N}}{1.2 \text{ kg} + 0.500 \text{ kg}} = \frac{4.905}{1.7} = 2.885 \text{ m/s}^2$$

Step 2 – Tension from m_1

Tension is the only sideways force on m_1, so it is m_1's net force:

$$\text{Mass 1:} \quad T = m_1 a = (1.2)(2.885) = 3.462 \text{ N}$$

FINAL ANSWER

a = 2.9 m/s^2 (m_1 toward the pulley, m_2 down)

T = 3.5 N

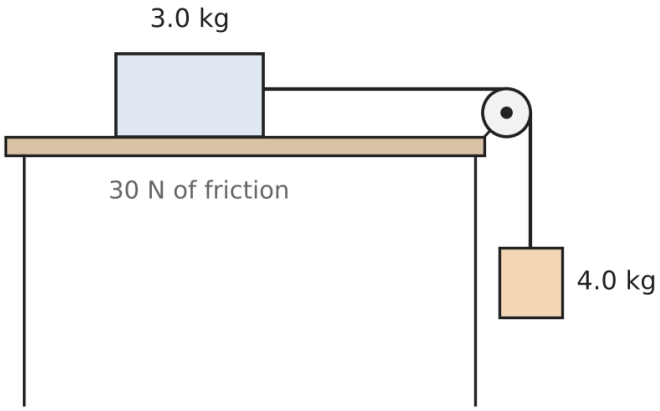
Check: mass 2: $F_{g2} - T = 4.905 - 3.462 = 1.443$ N = (0.500)(2.885).

TRY THESE 2

Tension without finding the acceleration

2. Find the tension in the string connecting the two blocks. Do not find the acceleration. There is 30 N of friction between the block and the table. The string is light and the pulley is frictionless. [34 N]

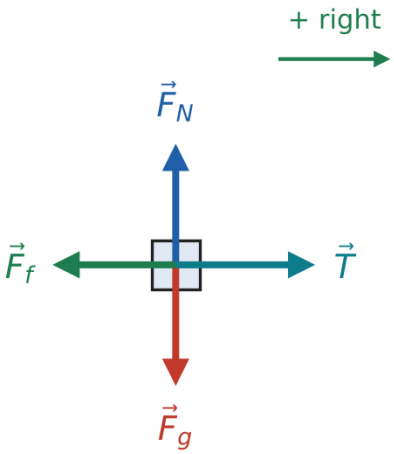
Picture the situation



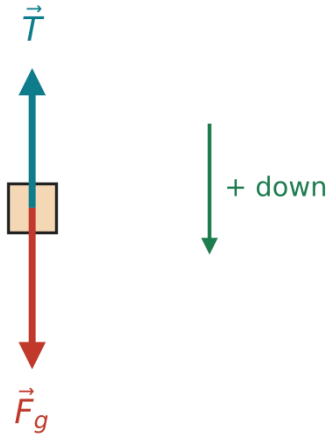
The 4.0 kg block falls and drags the 3.0 kg block across the table against 30 N of friction.

FBDs (+ for each block is the way it moves)

3.0 kg block



4.0 kg block



Step 1 – Newton's second law for each block

Both blocks have the same a, but we don't calculate it. Write one equation for each block:

Formula: $F_{net} = ma$

3.0 kg block (+ right): $T - F_f = (3.0)a \quad \Rightarrow \quad T - 30 = 3.0\,a$

Weight of the 4.0 kg block: $F_g = (4.0)(9.81) = 39.24$ N

4.0 kg block (+ down): $F_g - T = (4.0)a \quad \Rightarrow \quad 39.24 - T = 4.0\,a$

Step 2 – Eliminate a

Solve each equation for a. They are the same a, so set the two expressions equal:

Divide each by its mass: $a = \dfrac{T - 30}{3.0} \qquad a = \dfrac{39.24 - T}{4.0}$

Set them equal: $\dfrac{T - 30}{3.0} = \dfrac{39.24 - T}{4.0}$

Multiply both sides by 3.0 and by 4.0:

$4.0(T - 30) = 3.0(39.24 - T)$

Expand: $4.0\,T - 120 = 117.72 - 3.0\,T$

Add $3.0\,T$ and 120 to both sides: $7.0\,T = 237.72$

Divide both sides by 7.0 : $T = 33.96$ N

FINAL ANSWER

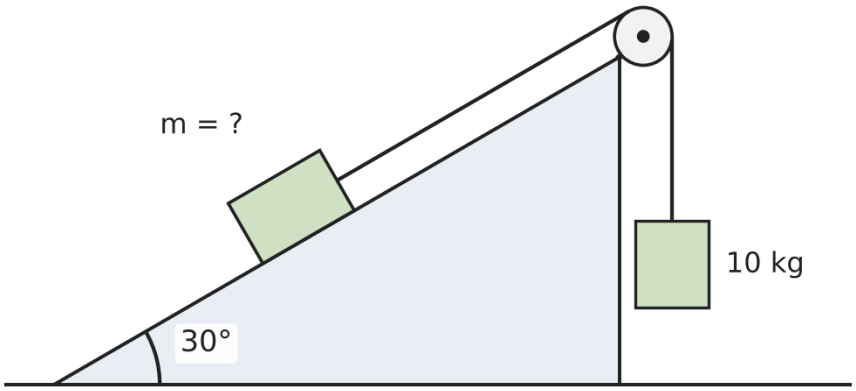
T = 34 N

TRY THESE 3

Finding the mass on the ramp

3. What is the value of *m* that would give the system an acceleration of 2.0 m/s/s? The coefficient of friction is 0.15, the string is light and the pulley is frictionless. [9.5 kg]

Picture the situation



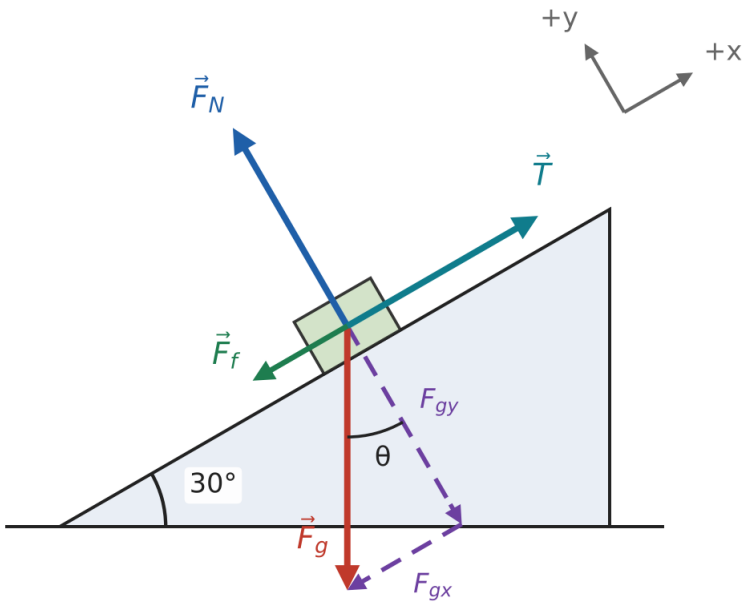
$\mu = 0.15,\ \vec{a} = 2.0\ \text{m/s}^2$

From the worksheet's diagram: a 30° ramp, with a 10 kg mass hanging from the rope.

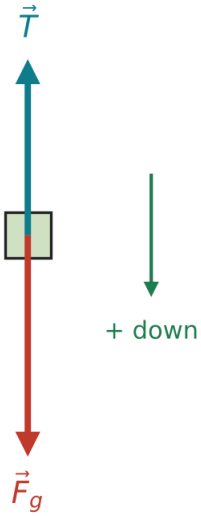
Take the 10 kg mass falling and pulling m up the ramp. Friction on m then points down the ramp. (The check at the end confirms this direction.)

FBDs (m: +x up the ramp, +y away from the ramp; 10 kg: + down)

m (sliding up the ramp)



10 kg (falling)



GIVEN

$a = 2.0\ \text{m/s}^2$	$\mu = 0.15$	$\theta = 30°$
$M = 10\ \text{kg}$	$T = ?$	$m = ?$

Step 1 – Tension from the hanging 10 kg mass

Formula: $F_{net} = ma$

10 kg mass (+ down): $F_g - T = Ma$

Subtract Ma and add T on both sides: $T = F_g - Ma$

$T = (10)(9.81) - (10)(2.0) = 98.1 - 20 = 78.1\ \text{N}$

Step 2 – Forces on m, in terms of m

F_{gx} (down the ramp) uses sine; F_{gy} (into the ramp) uses cosine and is balanced by F_N. Friction is μF_N:

$F_{gx} = mg\sin 30° = 4.9050\,m$

$F_N = F_{gy} = mg\cos 30° = 8.4957\,m$

$F_f = \mu F_N = (0.15)(8.4957\,m) = 1.2744\,m$

Step 3 – Newton's second law for m (+ up the ramp)

T points up the ramp (+). F_{gx} and F_f point down the ramp (−):

Mass m: $T - F_{gx} - F_f = ma$

Put in the expressions: $78.1 - 4.9050\,m - 1.2744\,m = 2.0\,m$

Add $4.9050\,m$ and $1.2744\,m$ to both sides:

$$78.1 = 2.0\,m + 4.9050\,m + 1.2744\,m$$

Combine the m terms: $78.1 = 8.1794\,m$

Divide both sides by 8.1794: $m = 9.548$ kg

FINAL ANSWER

m = 9.5 kg

Check: the direction is right: the hanging weight (98.1 N) is bigger than mg sin 30° + F_f = 46.8 + 12.2 = 59.0 N, so the 10 kg mass does fall.